

Spatial model of Real Estate portfolio exposure - Cholesky decomposition

$$Y(x) = ay(x) + bw(x) \int_0^T Z_w(sx) ds + Z_y(Tx),$$

where $Z_y(tx)$ is a partly time-changed Brownian sheet

$$Z_y(tx) = y \exp(y_1 x_1 + y_2 x_2) W_t^{v_y},$$

with the time changes

$$v_y^2(x_1) = \exp(2y_1 x_1), \quad v_y^3(x_2) = \exp(2y_2 x_2),$$

- deterministic component $ay(x)$, the integral of temperature deviations from the trend $\int_0^T Z_w(sx) ds$ weighed by a factor $bw(x)$, and a noise term $Z_y(Tx)$.

$$\text{Cov}[Z_y(tx)Z_y(sy)] = 2 \exp(y_1(t-s) + y_2(x_2 - y_2)).$$

$x_f = (x_1, \dots, x_d)$ are geographical positions. If the correlation matrix of $Z_y(tx_f)$ has the Cholesky decomposition

$$\text{corr}(Z_y(tx_f), Z_y(tx_k)) = \exp(y_1(x_{f1} - x_{k1}) + y_2(x_{f2} - x_{k2})),$$

then the vector $Z_f y(t)$ has the following representation

$$Z_f y(t) = Z_f y(0) + y W_t^y,$$

where W_t^y is a vector of independent standard Brownian motions. $w \in \mathbb{R}^{p+d}$ is the Cholesky decomposition of the spatial correlation matrix of $Z_w(tx)$ at locations $x_1, \dots, x_p$. Diagonal matrices

$$b_w = \text{diag}(b_w(x_f)), \quad h_w = \text{diag}(w(x_h)), \quad f_w = \text{diag}(w(x_f)),$$

RE thus follows a multivariate normal distribution with moments -

$$E[Y|F_t] = ay + bw \int_0^T Z_f w(s) ds + bw(1 - e^{w(T-t)})Z_f w(t) + Z_f y(t),$$

and

$$\text{Var}[Y|F_t] = \int_0^t (1 - e^{w(T-s)})^2 ds + bw f_w (h_w^T + f_w^T) f_w b_w + 2y(T-t)y^T y.$$

Agent Portfolios

Property Managers - portfolio holdings taken as of 2024 Financial Year, and weighted average extrapolated based on portfolio MV% (market value). The rent valuation time series is proxied by the valuation of nearest estate referenced from the Centaline Big Data transactions database.

The intersection criterion considered is a **ball tree algorithm**, which is a spatial indexing method utilizing **Harviside distance minimization** for spatial proximity. Inputs required are geographical coordinates (latitude and longitude) to create a hyper sphere.

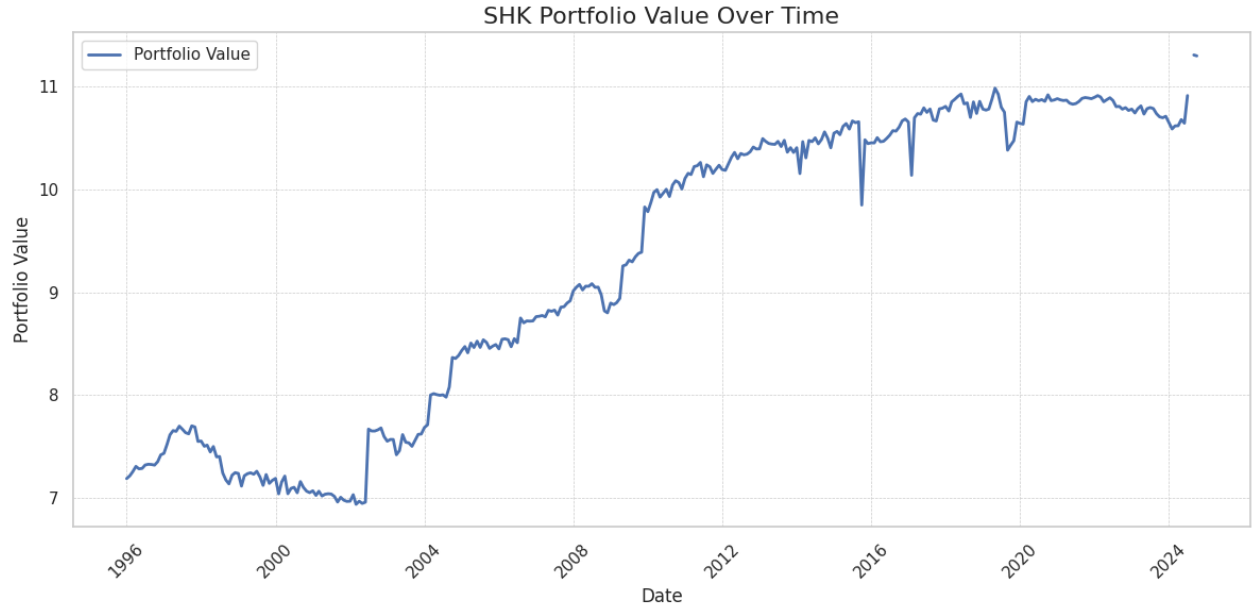


Figure 1: SHK Properties Manager - Time Series of their Hong Kong Portfolio's Rent Valuation. Portfolio as on 2024 FY

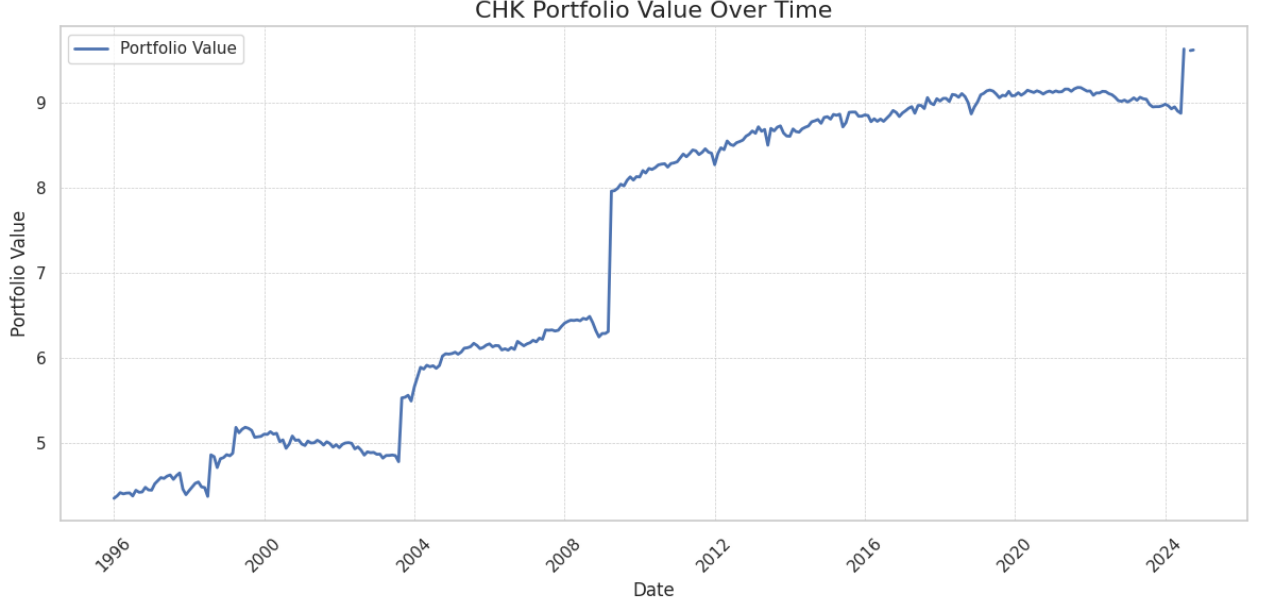


Figure 2: CHK Properties Manager - Time Series of their Hong Kong Portfolio's Rent Valuation. Portfolio as on 2024

Spatial and Temporal Effects Regression

The relationship between housing prices and climatic, temporal, and district characteristics can be expressed as follows:

$$\begin{aligned} \ln(\text{Portfolio rent valuation}_{i,t}) = & \beta_1 \ln(\text{temperature}_{i,t}) + \beta_2 \ln(\text{temperature}_{i,t}) \times \ln(\text{rainfall}_t) + \\ & \beta_3 (\text{number of severe typhoons}_t) \times (\text{number of coastal risk spots}_i) + \\ & \beta_4 \ln(\text{rainfall}_t) \times (\text{number of flooding blackspots}_{i,t}) \times (\text{climate news index}_t) \\ & + \delta(\text{other control variables})_{i,t} + \text{year effect} + \text{district fixed effect} + \epsilon_{i,t}. \end{aligned}$$

District Characteristics (Explaining district fixed effect) The district fixed effect accounts for cross-sectional variations in the following district specific characteristics:

- % built-up
- % vegetated
- % bare soil
- % rock
- % water

- % dense/high vegetation
- % sparse/low vegetation
- Building fraction (%)
- Road/impervious cover fraction (%)
- Average Sky View Factor (SVF)
- Average building height (m)
- Standard deviation of building height (m)

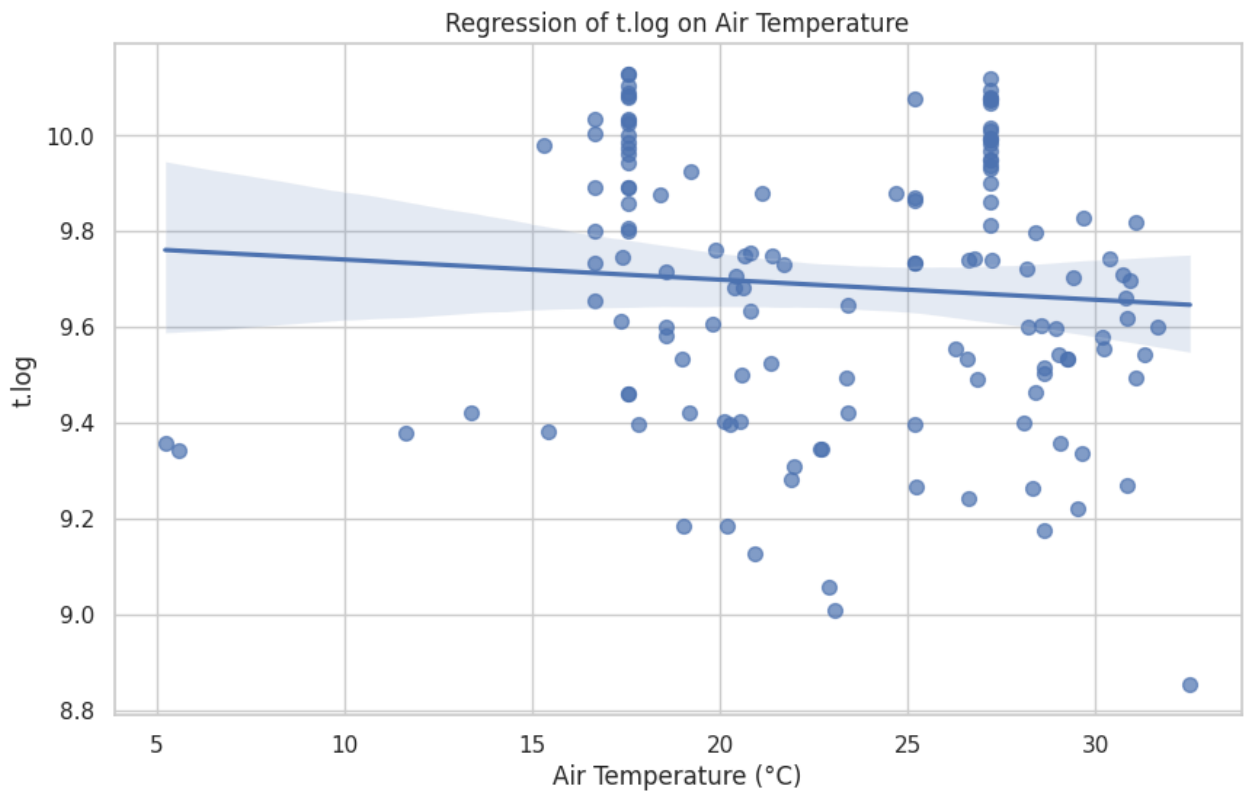


Figure 3: Univariate regression - Basic dependence of Log transformed rent valuation for an estate in **Kennedy Town** Ivy On Belcher' on the bi-hourly temperature time series of Station 6001 (**HK PolyU**). Polynomial method (degree = 2) interpolation used for missing temperature observations.